Stellar Cyber 6.6.0s is now live! Discover all the updates in the release notes!

Azure AD B2C SSO: Prepare Details for Policy / Key Configuration

The steps in this section are critical to perform first. Here, you obtain and note details for configuring the polices in the next section. You must also prepare the environment with certain keys and applications before you do that configuration.

If you stop and return to any of the procedures in this configuration process, ensure that you switch to the correct directory before you continue so that you are working on the correct Azure AD B2C Tenant container.

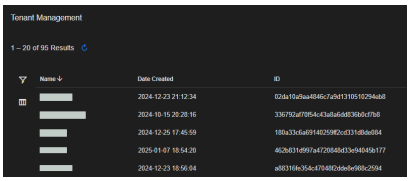
- **Before You Begin**
- **Set Azure AD B2C Directory & Obtain Tenant ID / Domain Name**
- **Create Azure AD B2C Signing and Encryption Keys**
- **Configure an IdentityExperienceFramework Application**
- **Configure a ProxyIdentityExperienceFramework Application**
- **Register the Stellar Cyber SAML Application**
- **Obtain IDs from the b2c-extensions-app**
- **Verify Preparations**

The order in which you perform certain steps matters. Perform the steps in exactly the order shown.

Before You Begin

- Make note of the FQDN of your Stellar Cyber Platform. You will be using for assorted configurations in this topic.
- Obtain the following from your Stellar Cyber configuration:
    - A server pem certificate and the private key for the server hosting your Stellar Cyber Platform. This will be uploaded in the Stellar Cyber Certificates screen.
    - Prepare a .pfx file which combines the certificate and the private key, for uploading to the Azure AD B2C server.

    IMPORTANT: Self-signed server certificates are not supported for configuration with Azure AD B2C.
- If you are configuring per-tenant SSO, obtain the Stellar Cyber ID of that tenant (**System | ORGANIZATION MANAGEMENT | Tenants**):



- The following worksheet may be helpful to track certain values that you will use during configuration, especially in the policy files. Note that the syntax / case matters for all of these values (see printable HTML version or Word version).

ATTRIBUTE	EXAMPLE	MYVALUES
Stellar Cyber Server Information (obtain before you begin)		
DP FQDN	testdp.stellarcyber.ai	

ATTRIBUTE	EXAMPLE



MYVALUES

ATTRIBUTE	EXAMPLE	MY VALUES
Tenant ID (for per tenant-SSO only)	29443942	
stellar_scope (authorization only)	root	
stellar_privilege (authorization only)	super_admin	
stellar_tenant (optional) (authorization only)	a112c31c04734b7ba5243e5e5432bfe6	
stellar_tenant_group (optional) (authorization only)	25320755	
Azure AD B2C Server Information (in order of occurrence)		
Domain / Primary Domain	stellarpmb2c.onmicrosoft.com	
Tenant ID / Tenant (Object ID)	04fcef19-20fe-4655-9498-3e5050b6dda1	
Application (Client) ID for IdentityExperienceFramework	01590824-a092-4271-99a3-f3e37b9f22cc	
Application (Client) ID for ProxyIdentityExperienceFramework	0c54849b-121f-496e-b1c3-406976bec48b	
SAML App Name	Stellar_SAML	
identifierUris	https://<Azure AD B2C tenant name>.onmicrosoft.com/<SAML application name> http://stellarpmb2c.onmicrosoft.com/Stellar_SAML	
Application ID URI	https://stellarpmb2c.b2clogin.com/stellarpmb2c.onmicrosoft.com	
b2c-extensions-app: Application (Client) ID	0c0abe38-79ab-45d3-99fe-9378eec7b16f	
b2c-extensions-app: Object ID	3c34826e-3bd5-47a2-b44d-08c09034cef2	

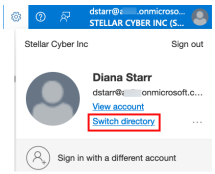
ATTRIBUTE	EXAMPLE	MY VALUES
Metadata URL	https://stellarpmb2c.b2clogin.com/stellarpmb2c.onmicrosoft.com/B2C_1A_signup_signin_saml/samlp/metadata	
Issuer URL	http://testdp.stellarcyber.ai	
replyUrlsWithType	Global SSO: https://testdp.stellarcyber.ai/saml/login/callback Per-tenant SSO: https://testdp.stellarcyber.ai/saml/login/callback/cust_id/a112c31c04734b7ba5243e5e5432bfe6	
Azure AD B2C User IDs (email & Issuer ID, if different)	SoCguy@stellarpmb2c.onmicrosoft.com	
	Example of issuer assigned ID that is not the same as email: AnalystTwo@stellarpmb2c.onmicrosoft.com, 6b744a55-9031-4b8c-ad6f-81dac5e74055@stellarpmb2c.onmicrosoft.com	

Set Azure AD B2C Directory & Obtain Tenant ID / Domain Name

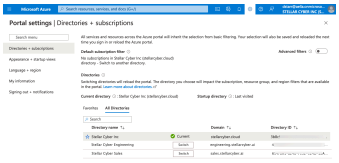
Steps

Ensure you are operating in the correct **Azure AD B2C** tenant area before you begin.

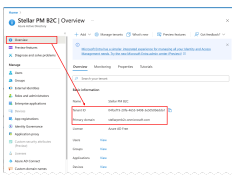
1. Sign in to the Azure portal.
2. Ensure you have selected the Azure AD B2C directory that contains the Microsoft Entra ID (formerly Azure AD) tenant that will be used to access Stellar Cyber. Navigate to the Directory selector using one of the following options:
    - Select the **Directories + subscriptions** icon in the portal toolbar.
    - Navigate to https://portal.azure.com/#settings/directory
    - Select the icon from your profile, then select **Switch**.



3. When the **Portal settings | Directories + subscriptions** page displays the list of directories, if there is only one directory listed, then no other step is required. If there is more than one directory, and the directory housing the tenant you are configuring for use with Stellar Cyber. is not currently active, click the **Switch** button on that row.



4. Now obtain the account identifying details that you will use throughout this configuration process. Navigate to the **Microsoft Entra ID > Overview** page.
5. Make note of the **Tenant ID** and **Primary Domain** and copy them to your worksheet for use in configuration later.



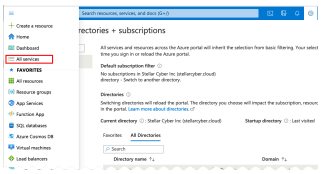
Create Azure AD B2C Signing and Encryption Keys

Steps

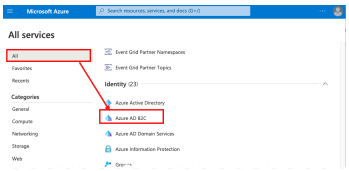
Generate two keys (one for signature and the other for encryption) in the Identity Experience Framework.

> If you stop and return to any of the procedures in this configuration process, ensure that you switch to the correct directory before you continue so that you are working on the correct Azure AD B2C Tenant container.

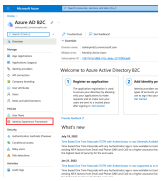
1. Access **All services** from the top-left corner of the Azure portal.



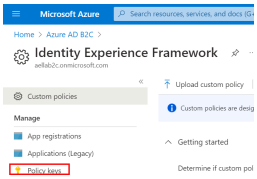
2. Search for and select **Azure AD B2C**.



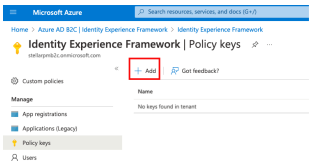
3. From the left navigation section for **Policies**, select **Identity Experience Framework.**



4. Select **Policy Keys** .



5. Select **Add**.



6. Create the first of two keys, the **Signing Key**, using the following settings:
    - **Options:** Select **Generate**
    - **Name**: Enter **TokenSigningKeyContainer**.
    - Activation date (optional)
    - Expiration date (optional)
    - **Key type**: Select **RSA**
    - **Key usage**: Enable **Signature**
7. Select **Create** to create the signing key which will be used in the Azure AD B2C custom policies later.

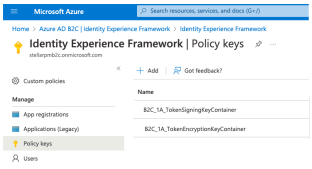
    The prefix **B2C_1A_**is automatically added to the name of your key.

    The Policy Keys page is refreshed and displays the key you created.
8. Select **Add** again to create the second key.
9. Create the **Encryption Key** using the following settings:
    - **Options**: Select **Generate**
    - **Name**: Enter **TokenEncryptionKeyContainer** . The prefix **B2C_1A_**is added automatically to the name of your key.
    - Activation date (optional)
    - Expiration date (optional)
    - **Key type**: Select **RSA**.
    - **Key usage**: Enable **Encryption**.

10. Select **Create** to create the encryption key which will be used in the Azure AD B2C custom policies later.

    The prefix **B2C_1A_**is automatically added to the name of your key.

    The Policy Keys page is refreshed and displays the key you created.



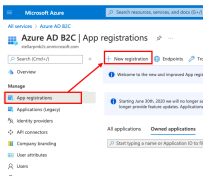
Configure an IdentityExperienceFramework Application

Steps

> If you stop and return to any of the procedures in this configuration process, ensure that you switch to the correct directory before you continue so that you are working on the correct Azure AD B2C Tenant container.

Create the IdentityExperienceFramework Application

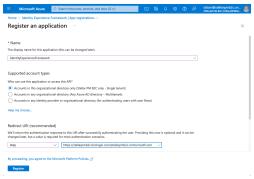
1. Navigate to **All services > Azure AD B2C > Select App registrations.**
2. Select **New registration**.



3. Create a new application with the following information:
    - **Name**: Enter **IdentityExperienceFramework**
    - **Supported account types**: Select **Accounts in this organizational directory only (Single tenant)**
    - **Redirect URI**: Set the following values:
        - **Web**
        - **https://<Azure AD B2C tenant name>.b2clogin.com/<Azure AD B2C tenant name>.onmicrosoft.com**

          where tenant name is your Azure AD B2C tenant domain name.

          Example: https://stellarpmb2c.b2clogin.com/stellarpmb2c.onmicrosoft.com
    - **Permissions**: select the **Grant admin consent to openid and offline_access permissions** check box



4. Select **Register**.

    The application is created.
5. Make note of the **Application (client) ID** on your worksheet.

    The ID will be used when you **Expose an API** and when you customize the **TrustFrameworkExtensions.xml** file later.

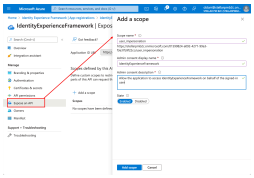
    Example value: Application (client) ID: 01590824-a092-4271-99a3-f3e37b9f22cc

Expose API for the IdentityExperienceFramework Application

1. With the **App registrations** page still displayed, select **Expose an API** from the **Manage** menu.
2. Select **Add a scope**.

    The dialog displays with an application ID URI. Do not modify this.
3. Select **Save and continue** to accept the default value.
4. Make note of the **Application ID URI** in your worksheet.
   Example: `https://stellarpmb2c.onmicrosoft.com/01590824-a092-4271-99a3-f3e37b9f22cc` for use later.
5. In the Scopes dialog, enter the following values to allow custom policy execution in your Azure AD B2C tenant:

- **Scope name**: Specify **user_impersonation**.
- **Admin consent display name**: Enter **Access IdentityExperienceFramework**.
- **Admin consent description**: Enter **Allow the application to access IdentityExperienceFramework on behalf of the signed-in user**.
- **State**:Ensure it is set to **Enabled**.



6. Select **Add scope**.

The IdentityExperienceFramework application is updated with the scope.

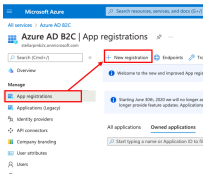
Configure a ProxyIdentityExperienceFramework Application

Steps

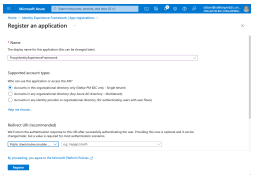
> If you stop and return to any of the procedures in this configuration process, ensure that you switch to the correct directory before you continue so that you are working on the correct Azure AD B2C Tenant container.

Create the ProxyIdentityExperienceFramework Application

1. Navigate to **All services > Azure AD B2C > Select App registrations**.
2. Select **New registration**.



3. Create a new application with the following information:
    - **Name**: Enter **ProxyIdentityExperienceFramework**
    - **Supported account types**: Select **Accounts in this organizational directory only (Single tenant)**.
    - **Redirect URI**: Set the following values:
        - **Public client/native (mobile & desktop)**
        - **myapp://auth**
    - **Permissions**: Select **Grant admin consent to openid and offline_access permissions**.



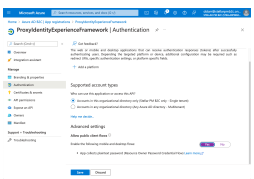
4. Select **Register**.

   The application is created.
5. Make note of the **Application (client) ID** on your worksheet. The ID will be used when you customize the **TrustFrameworkExtensions.xml** file later.

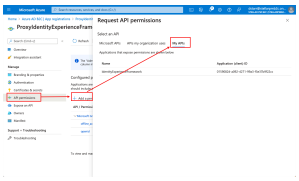
   Example value: Application (client) ID: 0c54849b-121f-496e-b1c3-406976bec48b

Configure the ProxyIdentityExperienceFramework Application

1. With the **ProxyIdentityExperienceFramework** application still displayed, select **Authentication** from the **Manage** menu.
2. In the dialog box that appears, locate the section for **Advanced settings > Allow public client flows**.
3. Toggle on **(Yes) Enable the following mobile and desktop flows**.



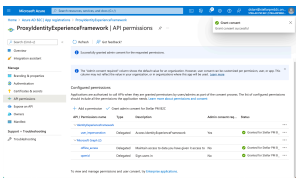
4. Select **Save**.
5. To verify that the setting is updated, select **Manifest** and locate the value `allowPublicClient` to check that the value is `true`.
6. While the **ProxyIdentityExperienceFramework** application is still selected, give the ProxyIdentityExperienceFramework access to the API you defined for the IdentityExperienceFrarmework Application by selecting **API permissions**.
7. Select **Add a permission**.
8. In the dialog box that opens, select the tab labeled **My APIs**.
9. Select the row with **IdentityExperienceFramework**.
10. In the dialog box that opens, select the **user_impersonation** permission you created earlier for the IdentityExperienceFramework.



11. Select **Add permission**.

    The API is added to the application but additional steps are required.
12. Select **Grant Admin consent** that is located next to **Add a permission**.
13. Select **Yes** to grant use of the API to the accounts in the current tenant.

    The API/Permissions status changes to **Granted**.

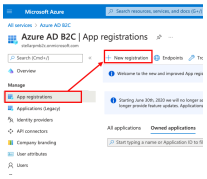


Register the Stellar Cyber SAML Application

Steps

If you stop and return to any of the procedures in this configuration process, ensure that you switch to the correct directory before you continue so that you are working on the correct Azure AD B2C Tenant container.

1. Navigate to **All Services > Azure AD B2C.**
2. Select **App registrations**, and then select **New registration**.



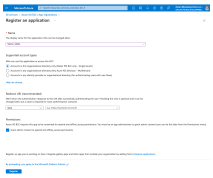
3. Use the following guidelines to configure the new application:

   Name: Enter **Stellar_SAML**.

   Supported account types: Select **Accounts in this organizational directory only**.

   Redirect URI: Specify **Web** and **https://localhost** (The URI will be updated below).

   Permissions: Leave **Grant admin consent to openid and offline_access permissions** selected.

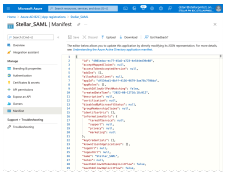


4. Select **Register**.

   The application details are displayed.



5. With the Stellar_SAML application still displayed, select the menu option for **Manage** and select **Manifest** to open the manifest editor for the application.



6. Locate and update (or add, if needed) the `identifierUris` attribute using the following syntax:

   `"identifierUris": [https://<your-Azure-AD-B2C-tenant-name>.onmicrosoft.com/<your-SAML-app-name>`

7. Locate and update (or add, if needed) the `samlMetadataUrl` attribute using the following syntax:

   - **If you are configuring global SSO**:

     `"samlMetadataUrl": "https://<your.stellar.cyber.address>/sso/saml/metadata"`

   - **If you are configuring per-tenant SSO**:

     `"samlMetadataUrl": "https://<your.stellar.cyber.address>/sso/saml/metadata/cust_id/<tenant-ID>"`

8. Locate and update (or add, if needed) the `replyUrlsWithType` attribute, using the following syntax:

   - **If you are configuring global SSO**:

     `"url": "https://<your.stellar.cyber.address>/saml/login/callback"`

   - **If you are configuring per-tenant SSO**:

     `"url": "https://<your.stellar.cyber.address>/saml/login/callback/cust_id/<tenant-ID>`

9. When you are finished, the fields in the manifest will look similar to the following samples:

   - **If you are configuring Global SSO**

```
{
    (snip)
    "identifierUris": [
        "https://stellarpmb2c.onmicrosoft.com/Stellar_SAML"
    ],
    (snip)
    "name": "Stellar_SAML",
    (snip)
    "publisherDomain": "stellarpmb2c.onmicrosoft.com",
    "replyUrlsWithType": [
        {
            "url": "https://testdp.stellarcyber.ai/saml/login/callback",
            "type": "Web"
        }
    ],
    (snip)
    "samlMetadataUrl": "https://testdp.stellarcyber.ai/sso/saml/metadata",
    (snip)
}
```

   - **If you are configuring Per-tenant SSO**

```
{
    (snip)
    "identifierUris": [
        "https://stellarpmb2c.onmicrosoft.com/Stellar_SAML"
    ],
    (snip)
    "name": "Stellar_SAML",
    (snip)
    "publisherDomain": "stellarpmb2c.onmicrosoft.com",
    "replyUrlsWithType": [
        {
            "url":
"https://testdp.stellarcyber.ai/saml/login/callback/cust_id/046ee23f3b4f4595858d925f451af980",
            "type": "Web"
        }
    ],
    (snip)
    "samlMetadataUrl":
"https://testdp.stellarcyber.ai/sso/saml/metadata/cust_id/046ee23f3b4f4595858d925f451af980",
    (snip)
}
```

If you have syntax issues, compare the file below:

Sample Manifest

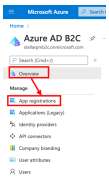

```
{
    "id": "aed035fa-1456-4a5a-b01a-e26817ac12e9",
    "acceptMappedClaims": null,
    "accessTokenAcceptedVersion": null,
    "addIns": [],
    "allowPublicClient": null,
    "appId": "74a51233-2158-410f-9ca7-809f27f0a8c7",
    "appRoles": [],
    "oauth2AllowUrlPathMatching": false,
    "createdDateTime": "2022-09-14T17:25:07Z",
    "description": null,
    "certification": null,
    "disabledByMicrosoftStatus": null,
    "groupMembershipClaims": null,
    "identifierUris": ["https://stellarpmb2c.onmicrosoft.com/Stellar_SAML"],
    "informationalUrls": {
        "termsOfService": null,
        "support": null,
        "privacy": null,
        "marketing": null
    },
    "keyCredentials": [],
    "knownClientApplications": [],
    "logoUrl": null,
    "logoutUrl": null,
    "name": "Stellar_SAML",
    "notes": null,
    "oauth2AllowIdTokenImplicitFlow": false,
    "oauth2AllowImplicitFlow": false,
    "oauth2Permissions": [],
    "oauth2RequirePostResponse": false,
    "optionalClaims": null,
    "orgRestrictions": [],
    "parentalControlSettings": {
        "countriesBlockedForMinors": [],
        "legalAgeGroupRule": "Allow"
    },
    "passwordCredentials": [],
    "preAuthorizedApplications": [],
    "publisherDomain": "stellarpmb2c.onmicrosoft.com",
    "replyUrlsWithType": [
        {
            "url": "https://testdp.stellarcyber.ai/saml/login/callback",
            "type": "Web"
        }
    ],
    "requiredResourceAccess": [
        {
            "resourceAppId": "00000003-0000-0000-c000-000000000000",
            "resourceAccess": [
                {
                    "id": "37f7f235-527c-4136-accd-4a02d197296e",
                    "type": "Scope"
                },
                {
                    "id": "7427e0e9-2fba-42fe-b0c0-848c9e6a8182",
                    "type": "Scope"
                }
            ]
        }
    ],
    "samlMetadataUrl": "https://testdp.stellarcyber.ai/sso/saml/metadata",
    "signInUrl": null,
    "signInAudience": "AzureADMyOrg",
    "tags": [],
    "tokenEncryptionKeyId": null
```


Obtain IDs from the b2c-extensions-app

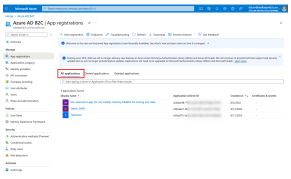
Steps

> If you stop and return to any of the procedures in this configuration process, ensure that you switch to the correct directory before you continue so that you are working on the correct Azure AD B2C Tenant container.

1. From **All Services > Azure AD B2C >Overview** screen, then select **App Registrations**.

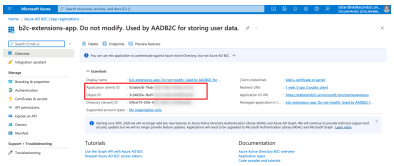


2. From the **App Registrations** pane, ensure that the **All Applications** sub-tab is selected.



3. Select the application named: **b2c-extensions-app. Do not modify. Used by AADB2C for storing user data**.

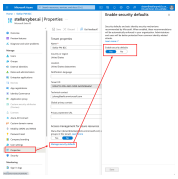
4. From the screen that displays, copy the **Application ID** and **Object ID** and save those to your worksheet. you will use these to modify the policy files later.



Enable Security Defaults for MFA (Optional)

If you plan to configure your environment to prompt for multi-factor authentication from Azure AD B2C, you must enable Security Defaults.

1. Browse to **Microsoft Entra ID > Properties.**
2. Select **Manage security defaults**.



3. Set the **Enable security defaults** toggle to **Yes**.
4. Select **Save**.

The remaining MFA configurations are accommodated in the sample templates provided in the next section.

Verify Preparations

Before you proceed to the next section, review your worksheet and ensure you have obtained information that you will use to configure the policies.

Proceed to the next step...

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